Global Renewable Energy in 2025:
How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix — spanning hydropower, wind, and solar technologies.

For: Policy Analysts & Energy Sector Strategists

Source: Our World in Data — Renewable Energy
Data: Ember (2026); Energy Institute – Statistical Review of World Energy (2025)

75% of Global GHG Emissions Still Come from Fossil Fuels

The urgency of the energy transition

75%

of global greenhouse gas emissions from fossil fuels

5M+

premature deaths per year from fossil-fuel air pollution

Since 1800s

Industrial Revolution locked in fossil fuel dominance

Since the Industrial Revolution, most countries' energy systems have been dominated by fossil fuels.

Burning fossil fuels for energy is the primary driver of the climate crisis and causes severe local air pollution — with at least 5 million premature deaths per year.

Rapidly scaling renewable energy — alongside nuclear — is essential to reduce both CO_2 emissions and air pollution and meet global climate targets.

Renewable energy includes solar, wind, hydropower, geothermal, bioenergy, wave, and tidal sources. Traditional biomass is excluded from modern renewable statistics.

Renewables Now Supply ~14% of Global Primary Energy

Primary energy = electricity + transport + heating combined (2024, substitution method)

~14%

of global primary energy
now from renewables

Why is 14% lower than the electricity share?

Transport and heating rely heavily on oil and gas, making them harder to decarbonise.

Regional leaders include Norway, Brazil, and New Zealand — deriving large shares of primary energy from renewables.

Fossil-fuel-producing nations in the Middle East and parts of Asia remain below 5%.

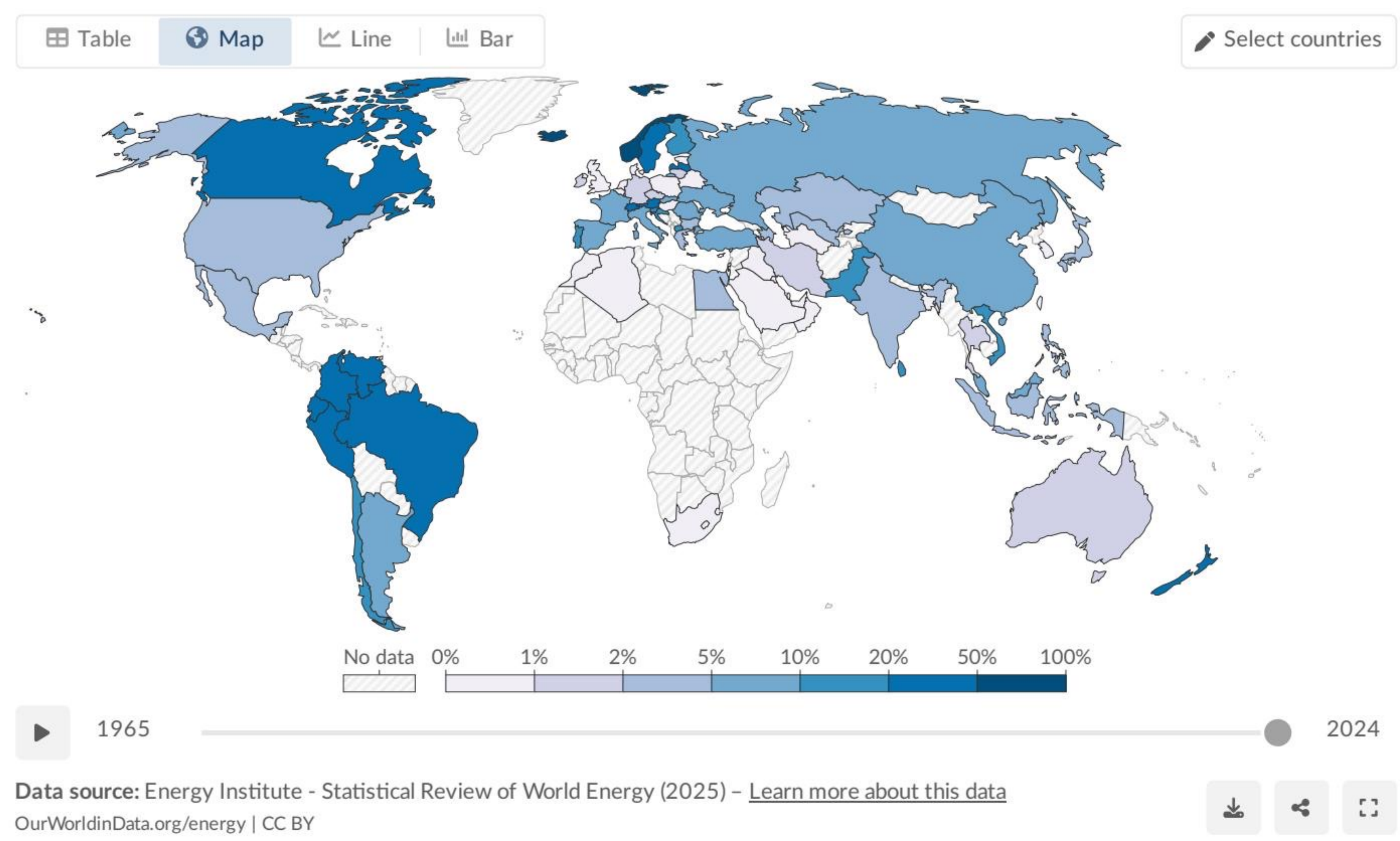
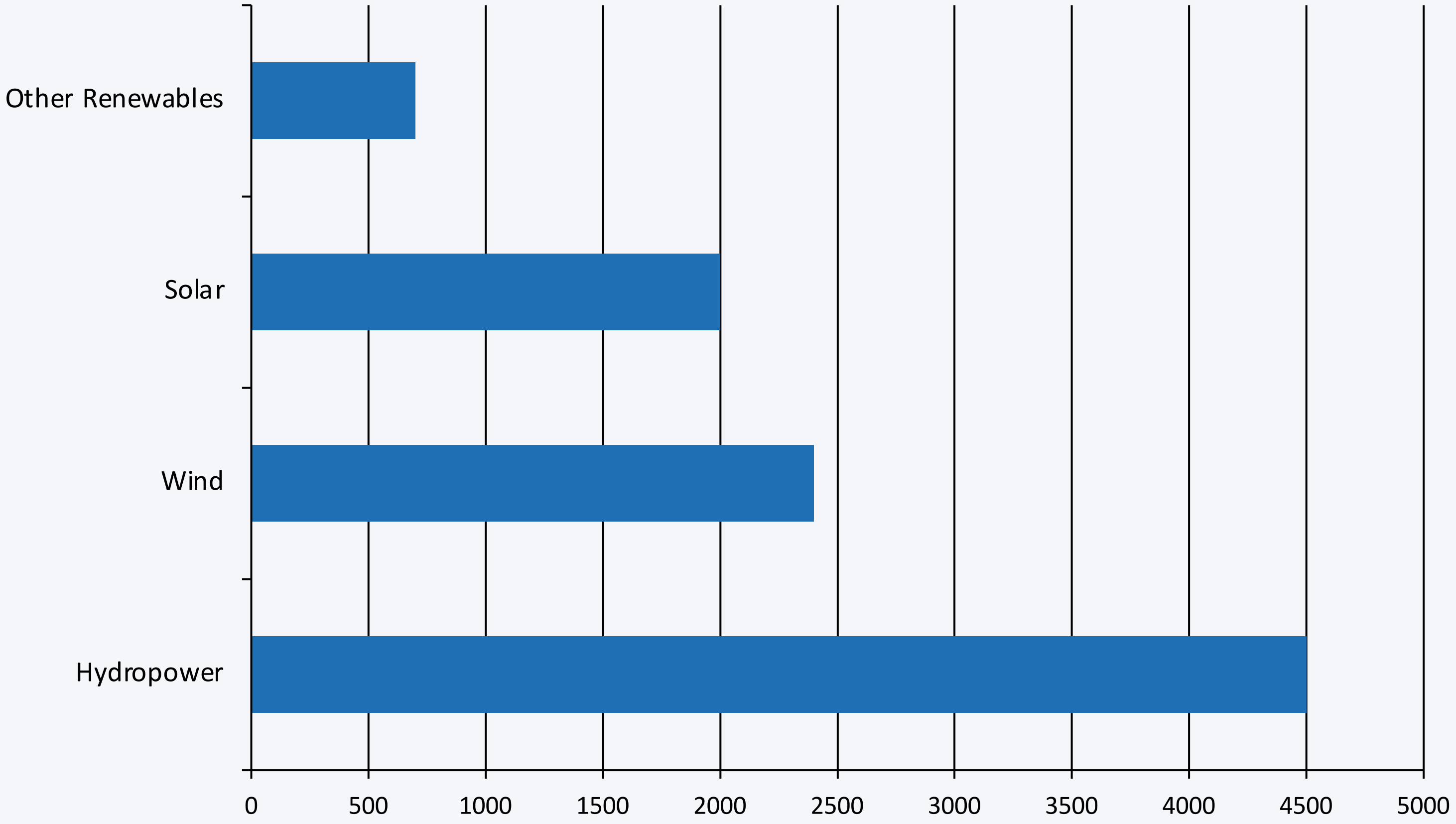


Figure: Share of primary energy from renewables by country, 2024 — Our World in Data

Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Global generation by technology, 2024 (approximate TWh)



Generation breakdown

Technology	~TWh (2024)	Share
Hydropower	~4,500	~47%
Wind	~2,400	~25%
Solar	~2,000	~21%
Other	~700	~7%

Key insight

Wind and solar combined (~4,400 TWh) now nearly match hydropower's output — and are growing far faster.

Solar is the fastest-growing energy technology in history, with dramatic expansion since 2010 driven by plummeting costs and strong policy support.

Source: Our World in Data — Renewable Energy (Ember 2026; Energy Institute 2025)

Almost One-Third of Global Electricity Now Comes from Renewables

Share of electricity from renewable sources by country, 2025

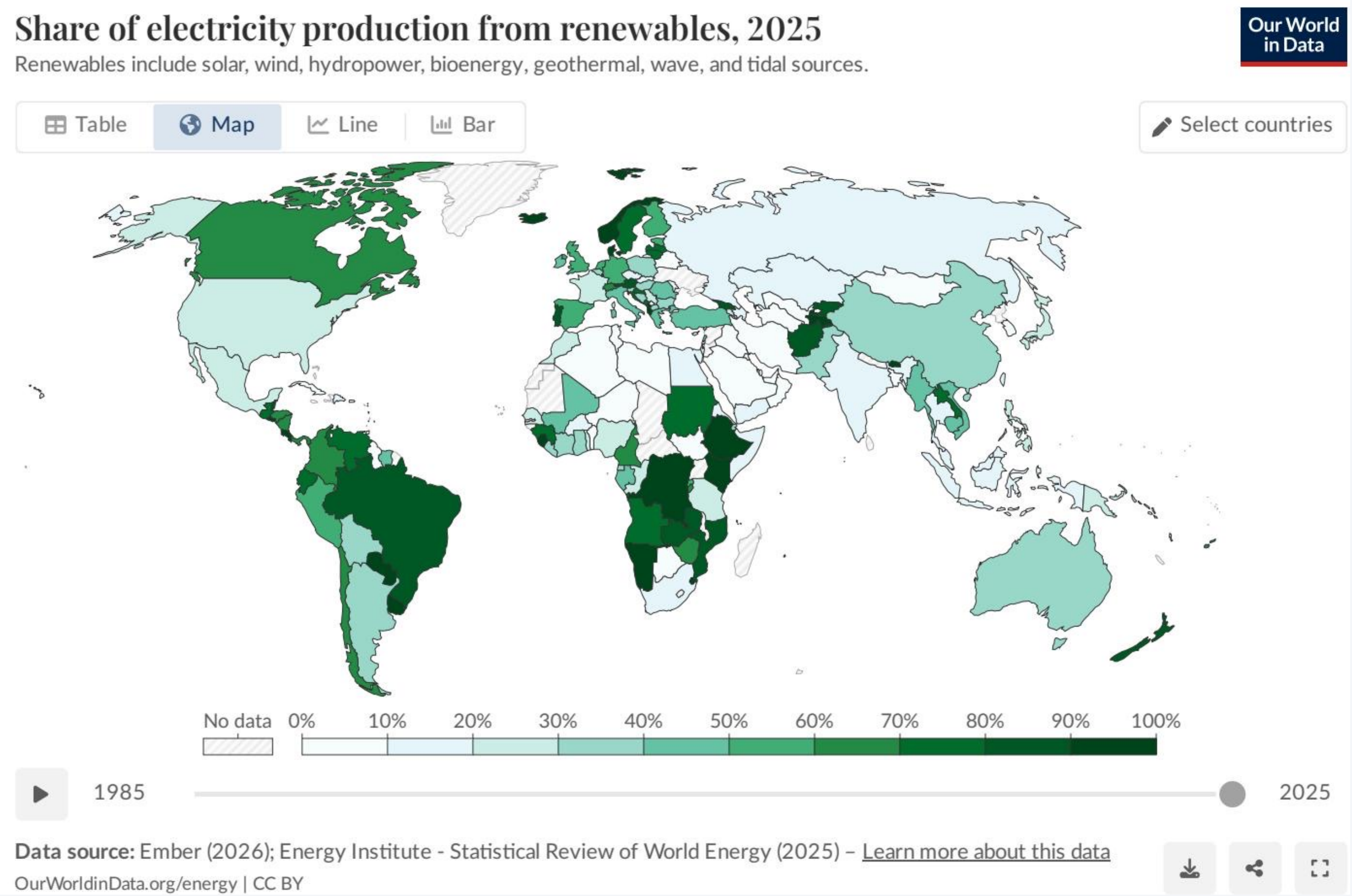


Figure: Share of electricity from renewables by country, 2025 — Our World in Data

Selected countries / regions

Country / Region	RE Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%
World avg.	~30%

Green = leaders (≥60%)

Amber = laggards (<10%)

Several African and Middle Eastern countries have renewable electricity shares below 10%, reflecting fossil fuel dependence and limited investment.

Hydropower Remains the Largest Renewable Source at ~50% of Generation

Hydropower has generated electricity at scale for over a century

~4,500 TWh

~47% of all renewable electricity generation (2024)

Top hydropower producers

1. China
2. Brazil
3. Canada

Growth potential is limited

While hydropower is the backbone of renewable electricity today, the best dam sites are largely already developed in many regions. Future growth will be driven primarily by wind and solar.

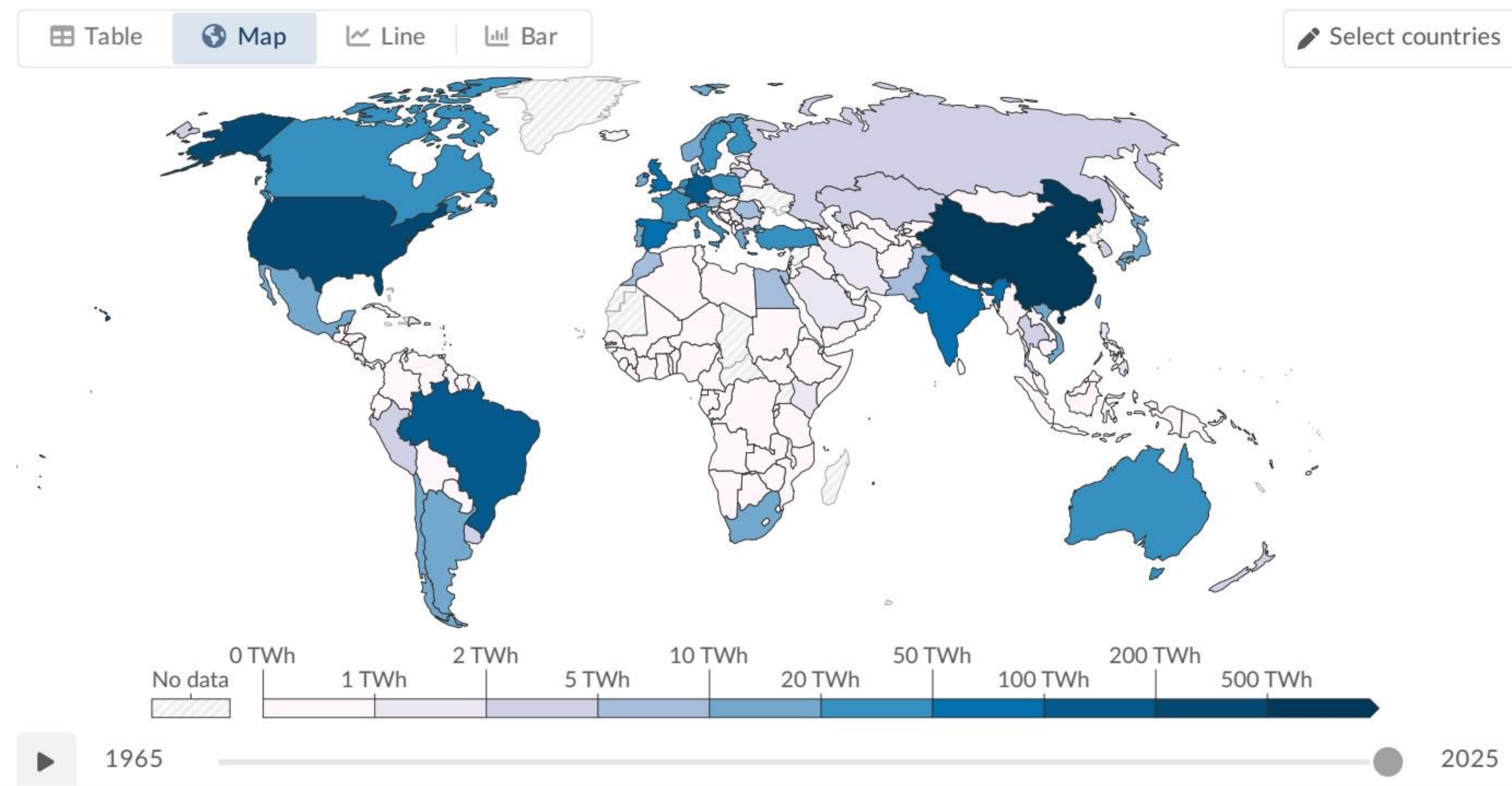


Figure: Hydropower generation by country, 2025 — Our World in Data

Wind and Solar Are the Fastest-Growing Energy Technologies in History

Wind ~2,400 TWh · Solar ~2,000 TWh · Combined nearly equals hydropower output

Wind

~2,400 TWh

~25% of renewable electricity
Onshore + offshore installations
Major regions: EU, China, USA

Solar

~2,000 TWh

~21% of renewable electricity
Fastest-growing since 2010
Driven by falling costs + policy

Wind + Solar combined: ~4,400 TWh — nearly matching hydropower's ~4,500 TWh output

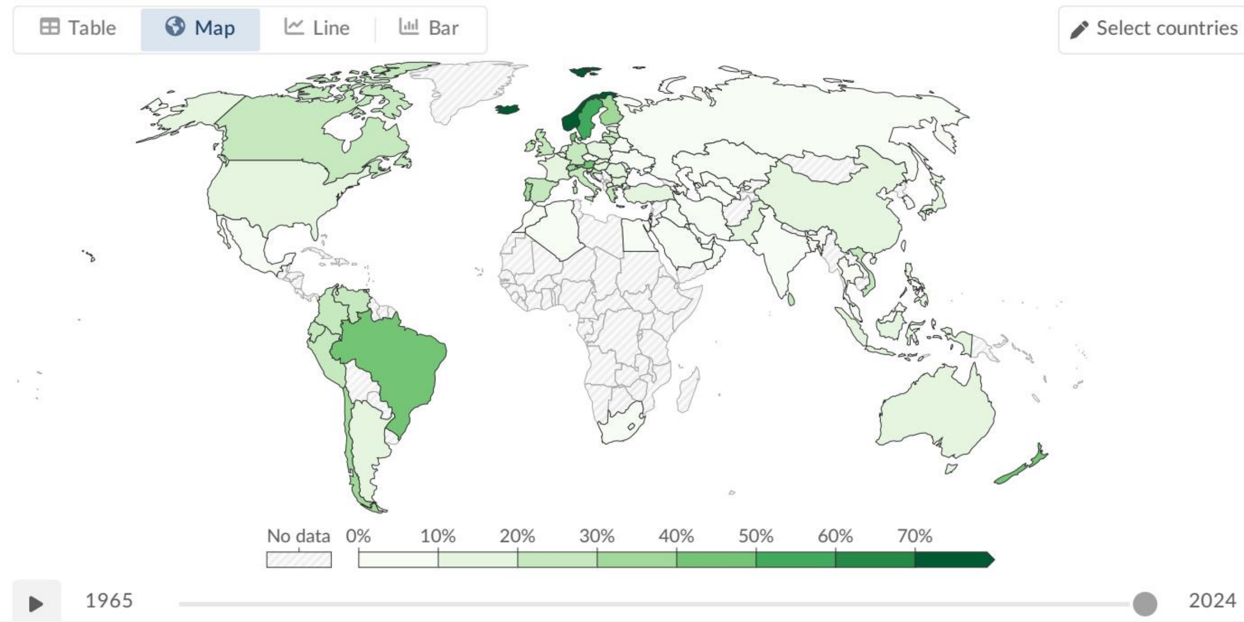


Figure: Wind power generation by country, 2025 — Our World in Data

Solar: fastest growth in energy history

Global solar generation has increased dramatically since 2010, driven by plummeting panel costs (down >90% over the decade) and strong government policy support worldwide.

Wind: major contributor in all leading economies

Wind generation has grown rapidly, with both onshore and offshore installations expanding significantly in Europe, China, and the United States.

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin America Lead

The renewable transition is deeply uneven across regions

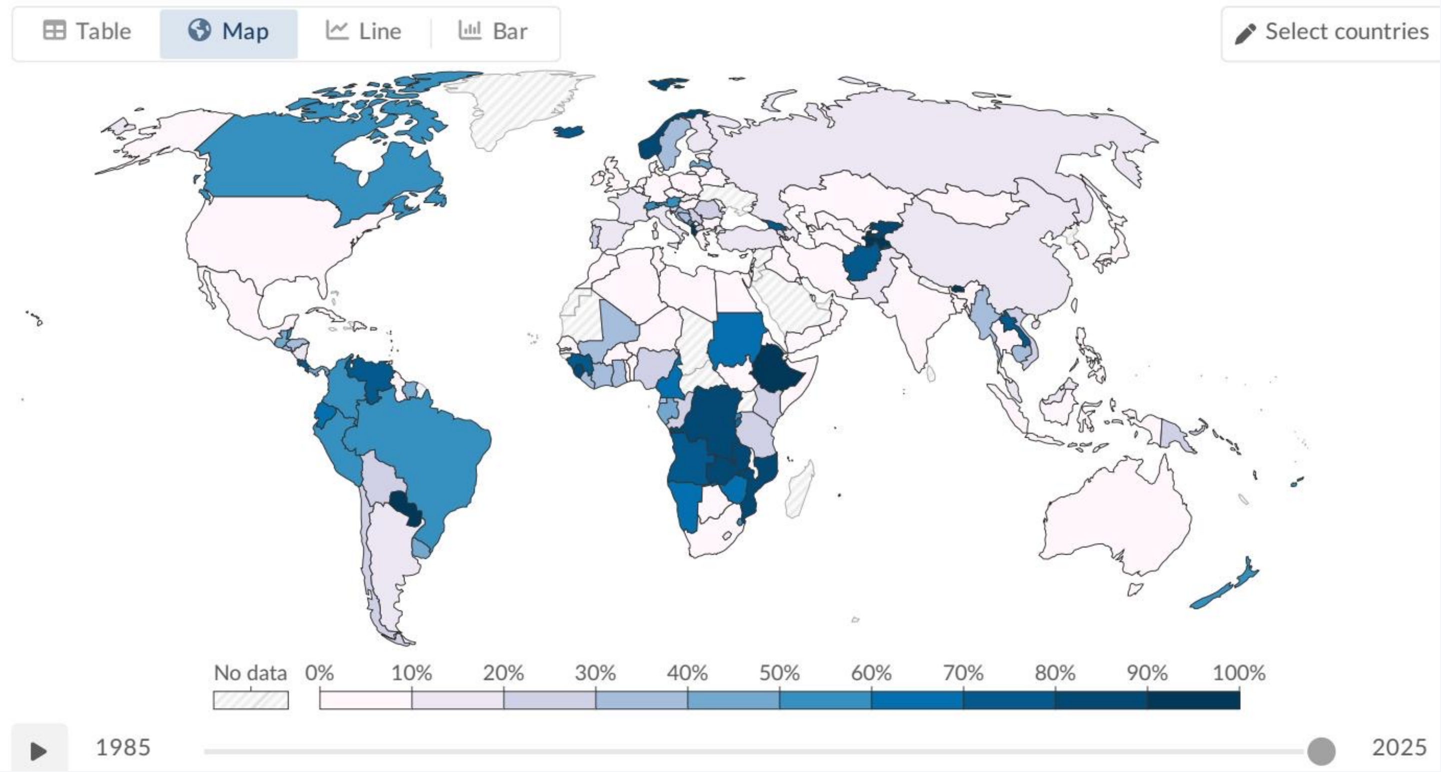


Figure: Share of electricity from hydropower by country, 2025 — Our World in Data

Regional snapshot

LEADERS

Norway / Scandinavia	**~98%+**	*Abundant hydropower + wind*
Brazil	**~85%**	*Hydro dominance + growing wind/solar*
Canada	**~65%**	*Large hydro fleet + wind expansion*

LAGGARDS

Middle East (e.g., Saudi Arabia)	**<2%**	*Near-total fossil fuel dependence*
Sub-Saharan Africa (many nations)	**<10%**	*Limited investment, minimal infrastructure*
South Africa	**~10%**	*Coal-dominant; solar growing slowly*

Key takeaway

Denmark and Germany exemplify a different path — heavy investment in wind and solar has lifted their renewable share well above the global average, demonstrating that policy-driven buildout can compensate for limited natural hydropower resources.

Source: Our World in Data — Renewable Energy (Ember 2026; Energy Institute 2025)

Key Takeaways: Renewables Are Growing Fast — but the Transition Must Accelerate

1 **~14% primary / ~30% electricity**

Renewables now supply ~14% of global primary energy and nearly one-third of global electricity — a significant milestone, but well short of what climate targets require.

2 **Hydro still dominates — but wind & solar are catching up**

Hydropower (~4,500 TWh, ~47% of renewable electricity) remains the backbone, but wind (~2,400 TWh) and solar (~2,000 TWh) together nearly match it and are growing far faster.

3 **Solar is the fastest-growing technology in energy history**

Solar costs have fallen >90% since 2010. Combined wind + solar now nearly equal all hydropower output.

4 **Stark regional disparities persist**

Scandinavia and Latin America exceed 65–98% renewable electricity; much of the Middle East and Sub-Saharan Africa remain below 10%.

5 **Transport & heating lag far behind electricity**

The overall primary energy share is suppressed by oil- and gas-dependent transport and heating. Electrification of these sectors is the next critical frontier.

Call to Action: Scale wind and solar deployment globally — especially in lagging regions — and accelerate electrification of transport and heating to close the gap between electricity and primary energy decarbonisation.

Source: Our World in Data — Renewable Energy | Data: Ember (2026); Energy Institute (2025)